# William Hermann

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Machine Learning Engineer passionate about democratizing AI through scalable, production-ready solutions. Experienced in developing specialized ML agentic pipelines. Proven track record in building production ready AI systems with Python.

## 1 Education

**Rijksuniversiteit Groningen** Groningen, NL

*BSc in Artificial Intelligence*

- **Relevant Coursework:** Machine Learning, Neural Networks, Deep Learning, Computer Vision, Data Structures & Algorithms, Agent Systems, Computational Linguistics.
- **Independent Research:** Experimented with large video model optimization using PyTorch and CUDA on university supercomputer clusters, focusing on efficient deployment strategies.

## 2 Core Competencies

**ML/AI Frameworks:** PyTorch, Transformers, CUDA, Hugging Face Libraries, Keras

**Languages:** Python, Rust, C++ (CUDA), TypeScript, SQL, Bash

**ML Infrastructure:** Supercomputer Clusters, Local Hardware Optimization, Production ML Systems

**Infrastructure:** Docker, Linux, Git, Matrix Server (Rust), CI/CD

**Deep Learning:** Computer Vision, NLP, Model Fine-tuning, Neural Architecture

**Systems:** Performance Optimization, Scalable Software Design, REST APIs

## 3 Professional Experience

**Lead Architect & Developer – Agentic AI Learning Platform** 2023 – Present

- **Architected** production-grade ML pipeline integrating LLMs with custom fine-tuning for personalized learning outcomes.
- Engineered high-performance system using **Python** backend with optimized inference serving for reduced latency.
- Implemented comprehensive monitoring and system health tracking for ML models in production environment.
- Built scalable infrastructure for concurrent real-time model inference using modern deployment practices.

**E-commerce & Client Technical Support (Independent)** 2021 – 2023

- Developed automated ML solutions for customer behavior prediction and personalization.
- Created data pipelines for training recommendation models using collaborative filtering techniques.

**Designer / Prototyper – Purple Mountain Innovations** 2018 – Mar 2022

*Contributed to rapid prototyping of ML-powered products in startup environment.*

## 4 Key ML/AI Projects

**Video Model Optimization Experiments**

- Explored CUDA kernel optimization for video transformers as a personal project, testing on both supercomputer clusters and local GPU.
- Experimented with model deployment strategies for resource-constrained environments.
- *Tech Stack: PyTorch, CUDA, Python, Transformers.*

**Natural Language to SQL Query Generator**

- Built production-ready system using Hugging Face Transformers, achieving high accuracy on domain-specific queries.
- Implemented efficient inference pipeline with model quantization and optimization techniques.

- *Tech Stack: Python, Hugging Face Transformers, PyTorch, ONNX, SQL.*

**Open-Source Contributions & Personal Infrastructure**

- Active contributor to ML libraries focusing on model optimization and industrial deployment.
- Maintain home lab infrastructure including self-hosted Matrix server written in Rust for secure communications.
- Developed utilities for model performance monitoring and system health tracking in production environments.

## 5 Certifications & Publications

- **Neo4j Certified Professional**
- **Upcoming:** "Efficient Video Model Deployment in Production Environments" - Under review